Chapter 4
Magnetic Fields and Magnetic Forces

Prof. Dr:
Mohamed Abdel-Rahman

Magnetic Dipoles: A current loop, of area A, behaves like a magnet and has a magnetic dipole moment:

$$\boldsymbol{M} = IA\hat{\boldsymbol{n}}$$

Where **n**ˆ is a unit vector normal to the plane of the loop. Its direction is found by placing the **fingers** of the right hand along the **current**; the **right thumb** then indicates the direction of $\hat{\mathbf{n}}$ and **M** . For N-current loops

$$\boldsymbol{M} = NIA\hat{\boldsymbol{n}} .$$

In a **uniform magnetic field**, a magnetic dipole behaves much like an electric dipole in uniform electric field, and the net force on the dipole will be zero. The torque, $\boldsymbol{\tau}$ on the dipole and the potential energy, U, are given by:

$$\tau = \boldsymbol{M} \times \boldsymbol{B}, \qquad U = MB\cos\theta$$

The magnetic torque on a flat current-carrying loop of wire by a uniform magnetic field **B** is maximum when the plane of the loop is parallel to **B** .

Example 9: The plane of area 4.0 cm^2 rectangular loop of wire is parallel to a 2.0 T magnetic field. The loop carries a current of 6.0 A. Calculate the magnitude of the torque acts on the loop.

Answer:

$$\tau = \boldsymbol{M} \times \boldsymbol{B} = NIA \times \boldsymbol{B} = NIAB\sin\theta$$

$$= (1)(6)(4\times10^{-4})(2)\sin 90^{\circ} = 4.8\times10^{-3}\ \text{N.m}$$

Example 11: A 100-turn circular coil of wire with radius 1.0 cm carries a current of 0.5 A. What torque will be exerted on the coil when it is placed in a magnetic field of 5.0 mT which makes an angle of 60° with the plane of the coil?

Answer:

$$\boldsymbol{M} = NI\boldsymbol{A} = (100)(0.5)\pi(1.0\times10^{-2})^2\ \hat{\mathrm{k}}$$

$$= 0.0157\ \hat{\mathrm{k}}.$$

$$\tau = \boldsymbol{M}\times\boldsymbol{B} = 0.0157\times5.0\times10^{-3}\sin 30^\circ = 3.93\times10^{-5}\ \mathrm{N.m}$$

Example 14: A current of 16 mA is maintained in a single circular loop of radius 0.32 m. An external magnetic field of 0.8 T is directed parallel to the plane of the loop.

a- Calculate the magnetic moment of the current loop.

Answer:

$$|\boldsymbol{M}| = IA = 16\times10^{-3}\times\pi(0.32)^2$$

$$= \underline{5.1\times10^{-3}\ \mathrm{C\cdot m^2}}$$

b- What is the magnitude of the torque exerted on the loop by the magnetic field?

$$|\tau| = |\boldsymbol{M}\times\boldsymbol{B}| = |\boldsymbol{M}||\boldsymbol{B}|\sin 90^\circ = 5.1\times10^{-3}\times0.8$$

$$= \underline{4.1\times10^{-3}\mathrm{C\cdot m^2T}}$$

Sources of Magnetic Field

1. Magnetic field of a moving charge:

To find the magnetic field B of a point charge q moving with constant velocity v at the point P, let's call the location of the moving charge at a given instant the **source point** and the point P where we want to find the field the **field point**. Experiments show that the magnitude of B is proportional to q, the particle velocity v, $1/r^2$ and to the sinφ . But the direction of B is perpendicular to the plane containing the line from source point to field point and the particle velocity v.

$$B = \frac{\mu_0}{4\pi}\frac{|q|\, v\sin\varphi}{r^2} \qquad \vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$$

Where $\mu_0/4\pi$ is proportionality constant, B has the greatest magnitude when sin φ = 1 or φ = 90 . If the charge q is negative, the direction of B is opposite. The unit of B is tesla (T).

2. Magnetic Field of a Current

The principle of superposition of magnetic fields: The total magnetic field caused by several moving charges is the vector sum of the fields caused by the individual charges.

We begin by calculating the magnetic field caused by a short segment dl of a current carrying a conductor, the total moving charge dQ in the segment is:

$$dQ = nqAdl$$

Where n is the number of the moving charged particles per unit volume, A is the cross sectional area, and Adl is the volume.

The magnitude of the resulting field dB at any point P is

$$dB = \frac{\mu_0}{4\pi}\frac{|dQ|\, v_d \sin\varphi}{r^2} = \frac{\mu_0}{4\pi}\frac{|dq|\, n v_d A dl \sin\varphi}{r^2},$$

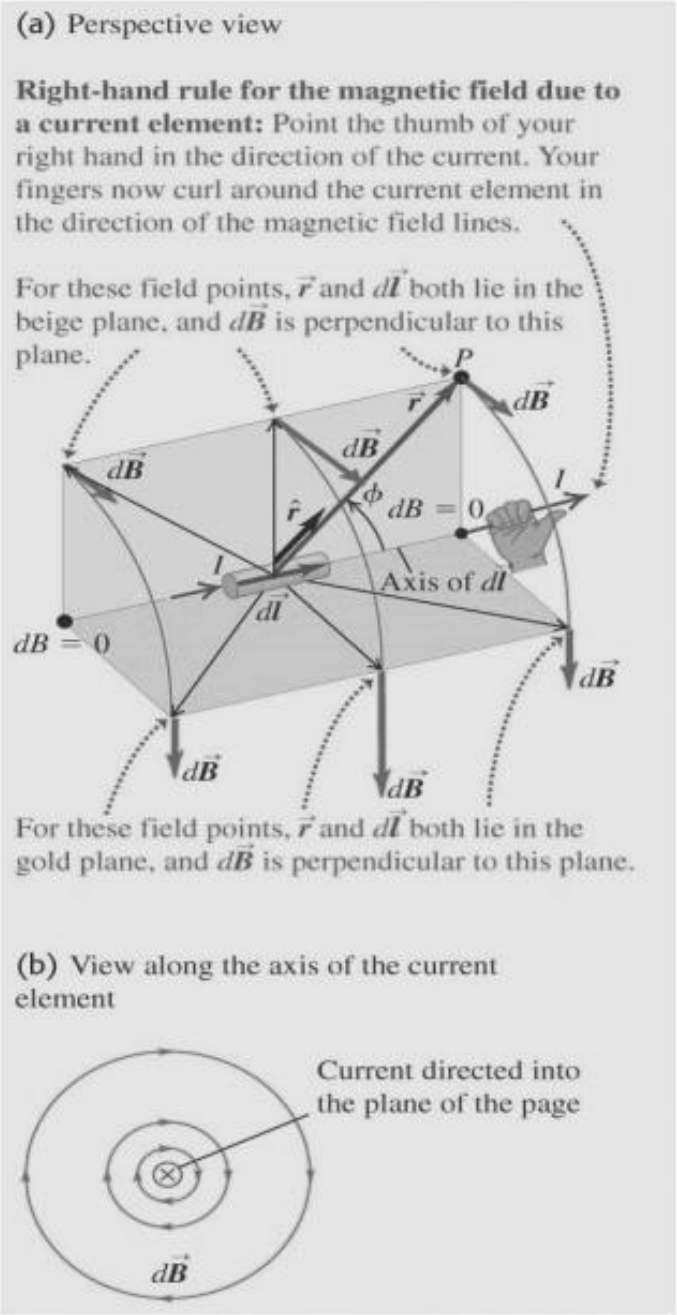
but $$n|dq|\, v_d A = I,$$

Then $$dB = \frac{\mu_0}{4\pi}\frac{I dl \sin\varphi}{r^2}$$

In the vector form, using the unit vector r^, we have

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I d\vec{l} \times \hat{r}}{r^2} \qquad \vec{B} = \frac{\mu_0}{4\pi}\int \frac{I d\vec{l} \times \hat{r}}{r^2}$$

This is the **Biot and Savart law**.



We can use this law to find the total magnetic field at any point in space due to the current in a complete circuit.

3. Magnetic Field of a Straight Current-Carrying Conductor

An important application of the Biot and Savart law is finding the magnetic field produced by a straight current carrying conductor.

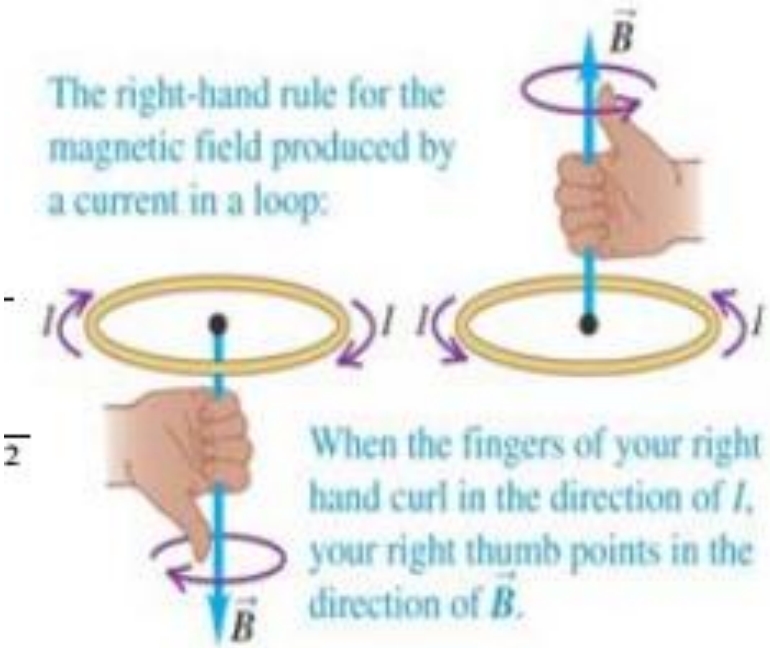
At all points on a circle of radius r around the conductor, the magnitude of B is:

$$B = \frac{\mu_0 I}{2\pi r}$$ (near a long, straight, current-carrying conductor)

Note:

The total magnetic flux through any closed surface is always zero. This implies that there are no isolated magnetic charges or magnetic monopoles. Any magnetic field lines that enters a closed surface must also emerge from that surface.

$$\oint \vec{B}.d\vec{A} = 0$$



4. Force Between Parallel Conductor The lower conductor produces a B field that, at the position of the upper conductor, has magnitude:

$$B = \frac{\mu_0 I}{2\pi r}$$

The force that this field exerts on a length L of the upper conductor is:

$$\vec{F} = I' \vec{L} \times \vec{B}$$

Where the vector L is in the direction of the current I ' and has the magnitude L. Since B is perpendicular to the length of the conductor and hence to L , the magnitude of this force is:

$$F = I' LB = \frac{\mu_0 I I' L}{2\pi r}$$

And the force per unit length is:

$$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}$$

Applying the right-hand rule to $\vec{F} = I' \vec{L} \times \vec{B}$ shows that the force on the upper conductor is directed downward.

Note:

1- The two conductors carrying current in the same direction attract each other. If the direction of either current is reversed, the forces also reverse.

2- Parallel conductors carrying currents in opposite directions repel each other.

5. Magnetic Field of a Circular Current Loop

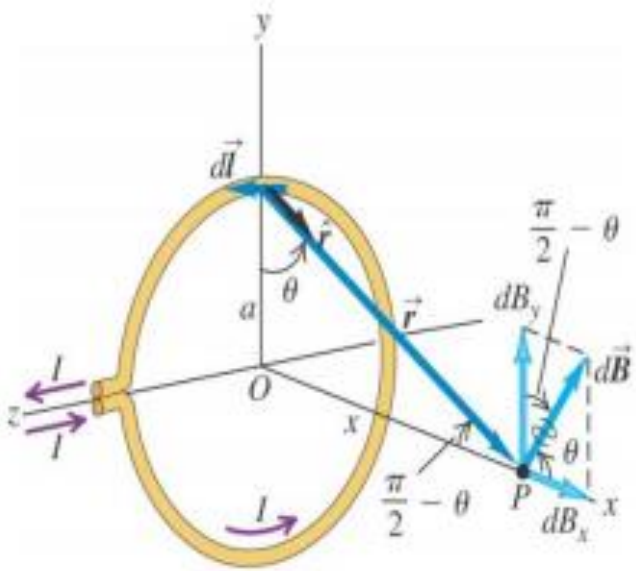
We can use the law of Biot and Savart to find the magnetic field at a point P on the axis of the loop, at a distance x from the center. As the figure shows, dl and r are perpendicular, and the direction of the field dB lies in the x-y plane.

$$\because r^2 = x^2 + a^2$$

$$\because dB = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)}$$

$$dB_x = dB\cos\theta = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \frac{a}{(x^2 + a^2)^{1/2}}$$

$$dB_y = dB\sin\theta = \frac{\mu_0 I}{4\pi} \frac{dl}{(x^2 + a^2)} \frac{x}{(x^2 + a^2)^{1/2}}$$



Since

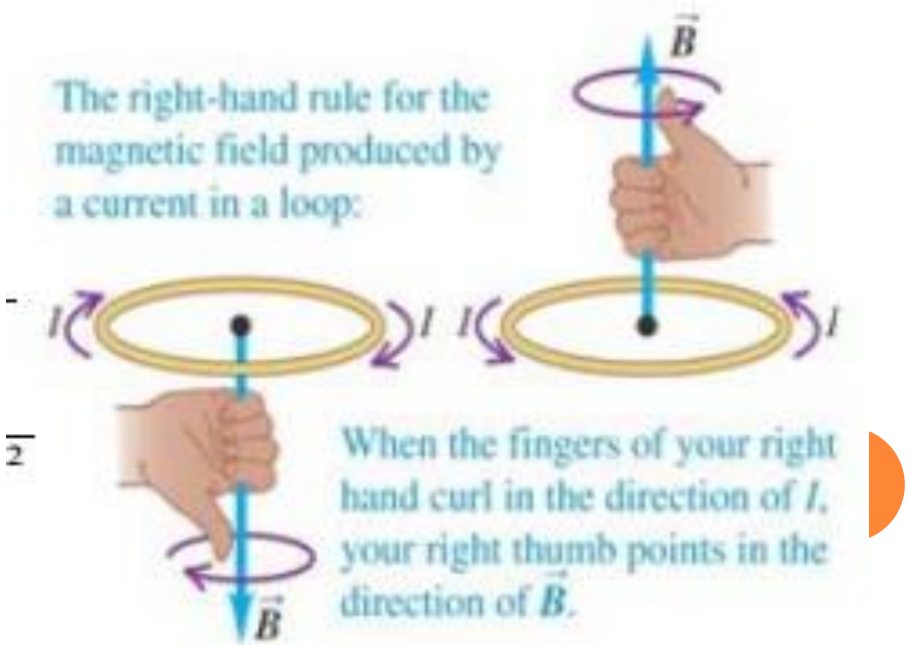
The situation has rotational symmetry about the x-axis, so there can't be a component of the total field B perpendicular to this axis. For every element d l there is a corresponding element on the opposite side of the loop, with opposite direction. These two elements give equal contributions to the x-component of d B , but opposite components perpendicular to the x-axis. Thus all the perpendicular components cancel and only the x-component survive.

$$B_x = \int \frac{\mu_0 I}{4\pi} \frac{a\,dl}{(x^2+a^2)^{3/2}} = \frac{\mu_0 I a}{4\pi (x^2+a^2)^{3/2}} \int dl$$

The integral of dl is just the circumference of the circle $\int dl = 2\pi a$ then:

$$B_x = \frac{\mu_0 I a^2}{2(x^2+a^2)^{3/2}}$$

The direction of the magnetic field is given by a right-hand rule.

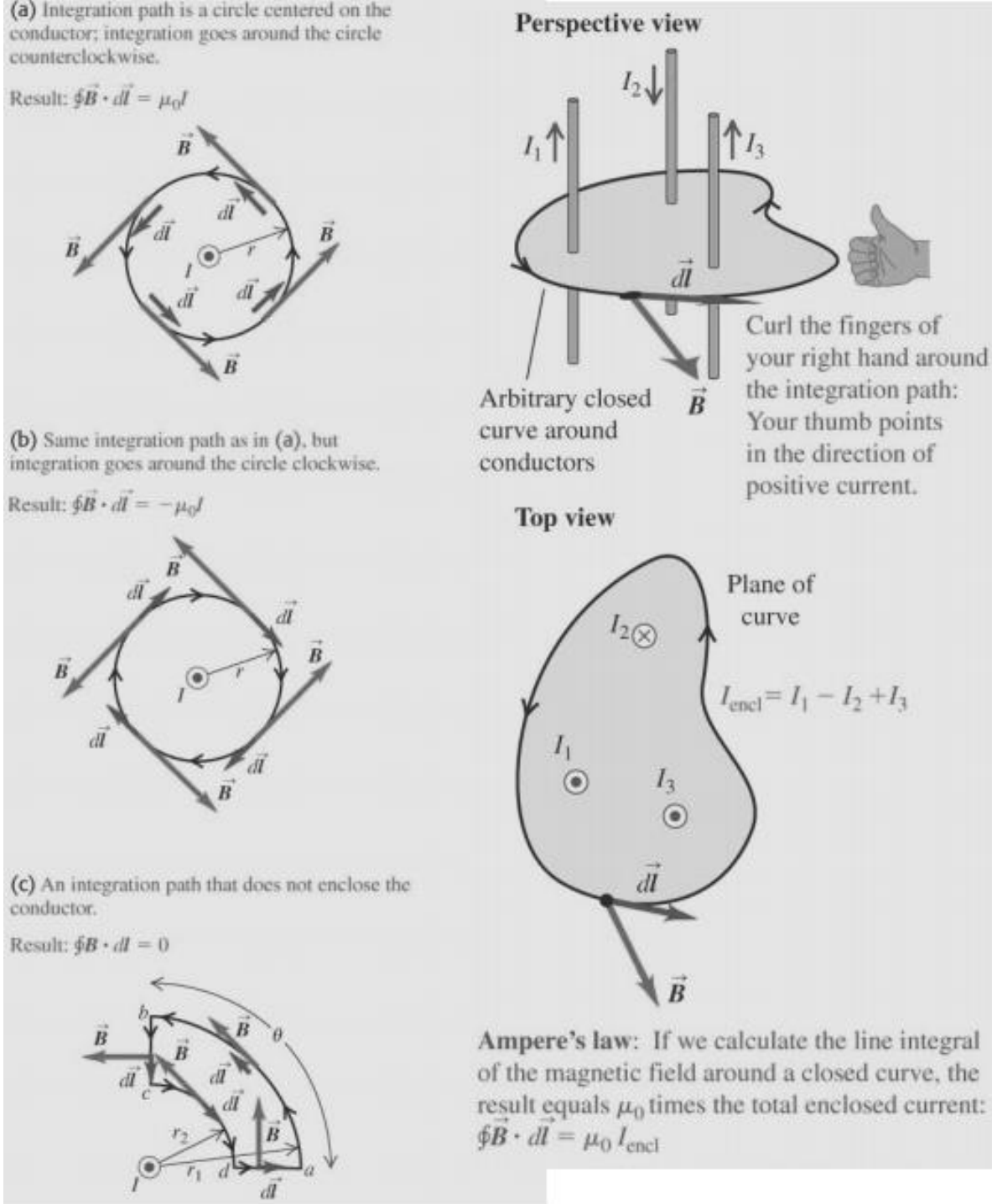


Ampere's Law

It states that: the line integral of B around any closed path equal to μ_0 times the net current through the area enclosed by the path.

$$\oint \vec{B}.d\vec{l} = \mu_0 I_{encl}$$

The positive sense of current is determined by a right-hand rule



Applications of Ampere's Law

Example 1: Magnetic field of a long, straight, current-carrying conductor:

$$\overbrace{B\oint ds = B(2\pi r) \qquad \mu_o i_{enc}}^{\oint \vec{B}.d\vec{s}}$$

$$\Rightarrow \boxed{B = \mu_o \frac{i_{enc}}{2\pi r}}$$

Example 2: Magnetic field **inside** a long straight wire with current (a long cylindrical conductor): a cylindrical conductor with radius R carries a current. The current is uniformly distributed over the cross sectional area of the conductor.

$$\overbrace{B\oint ds = B(2\pi r) \qquad \mu_o i_{enc} = \mu_o i \frac{\pi r^2}{\pi R^2}}^{\oint \vec{B}.d\vec{s}}$$

$$\Rightarrow \boxed{B = \frac{\mu_o i}{2\pi R^2} r} \qquad r \prec R$$

Current density (current per unit area) $\qquad J = \dfrac{i}{\pi R^2}$, so

$$i_{enc} = J(\pi r^2) = \frac{i\pi r^2}{\pi R^2}$$

Magnetic field of a solenoid:

a coil of wire that acts as a magnet when an electric current flows through it

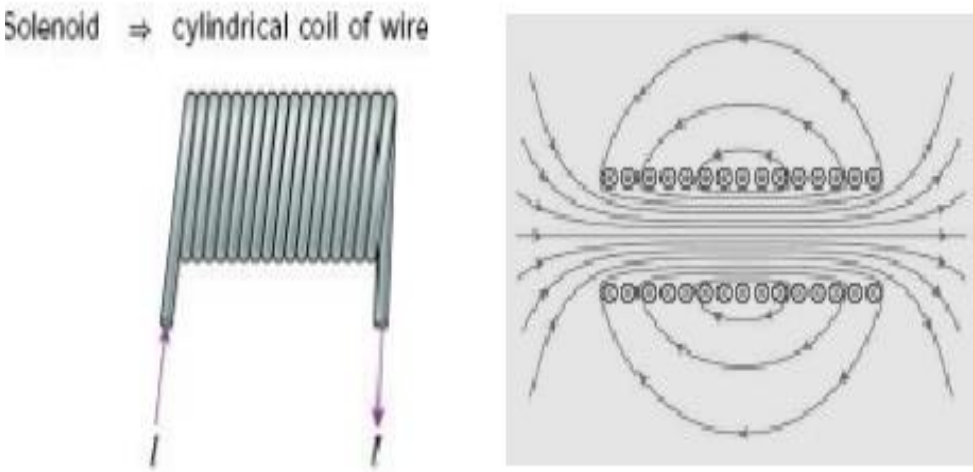
$$B_s = \mu_o nI = \mu_o \left(\frac{N}{L}\right) I$$

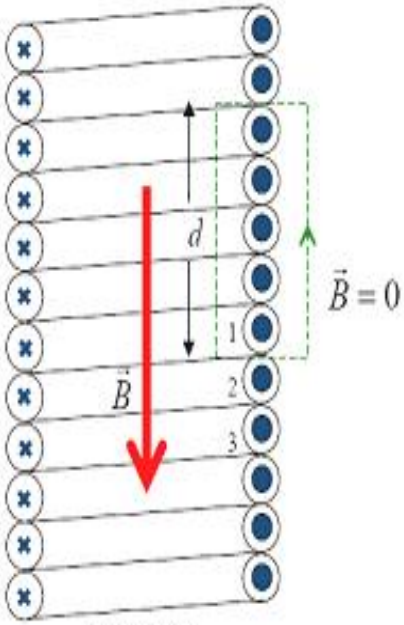
inside solenoid, near centre

Where B is the magnetic field anywhere **within** the solenoid, its unit is Tesla (T), μ_0 is the permeability of free space, $\mu_0 = 4\pi \times 10^{-7}$ T.m/A , n is the number of coils per meter in the solenoid, L is the length of the solenoid, and I is the current passing through the solenoid.

B ≈0

outside solenoid





A solenoid